



NAME: _____

TEACHER: _____

GOSFORD HIGH SCHOOL

2012/2013

EXTENSION 1 MATHEMATICS

HSC ASSESSMENT TASK 1.

Time Allowed: 60 minutes (plus 5 min. reading time)

- Write using black or blue pen.
- Board-approved calculators may be used.
- Sections 3 and 4 should be started on a new page.
- All necessary working should be shown in Section 2 , 3 and 4

SECTION	QUESTION TYPE	MARKS	RESULT
1	MULTIPLE CHOICE	4	
2	PARABOLA	13	
3	INDUCTION	8	
4	FURTHER TRIG	18	
	TOTAL	43	

MATHEMATICS
Extension 1
Assessment Task 1
December 2012

Time: 1 hour plus 5 minutes reading time

SECTION 1

Multiple choice.

(Write your answer on your answer sheet: **NOT** on the test sheet)

- 1) The length of the latus rectum of the parabola $y = x^2$ is:

A) $\frac{1}{4}$

B) 1

C) 2

D) 4

- 2) A correct expression for $\cos 4A$ is:

A) $2\cos^2 A - 1$

B) $2\cos^2 2A - 2$

C) $1 - 2\sin^2 A$

D) $2\cos^2 2A - 1$

- 3) The equation of the chord of contact from the point (1,-1) to the parabola $x^2 = 8y$ is:

A) $x - 4y + 4 = 0$

B) $x - 2y + 2 = 0$

C) $x - 8y + 8 = 0$

D) $x + 4y - 4 = 0$

- 4) When written in terms of t , where $t = \tan \frac{\alpha}{2}$, $\frac{1 + \sin \alpha + \cos \alpha}{1 + \sin \alpha - \cos \alpha} \equiv$

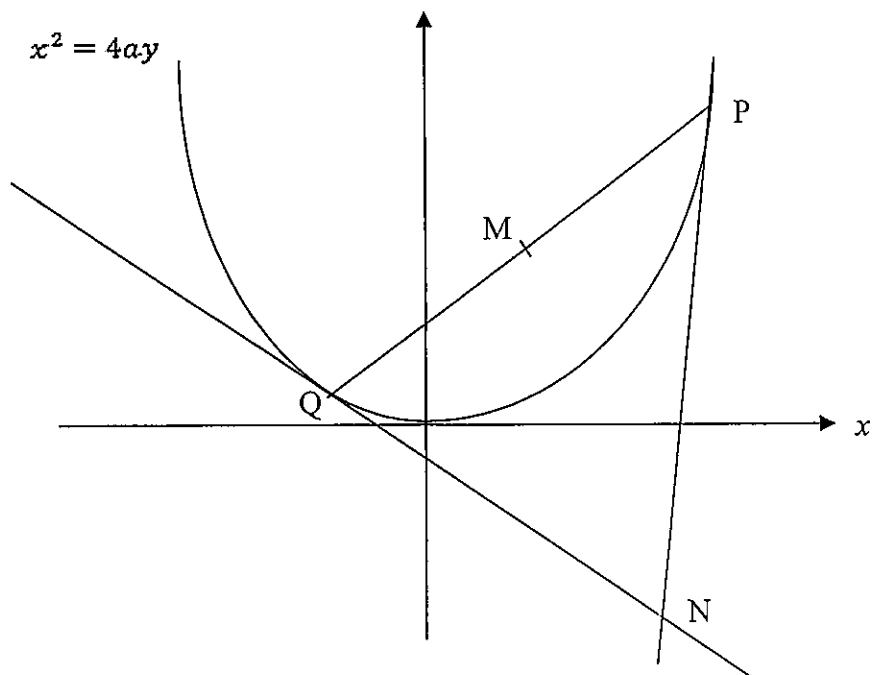
A) t

B) $-t$

C) $\frac{1}{t}$

D) $-\frac{1}{t}$

SECTION 2



1) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two points on the parabola $x^2 = 4ay$. M is the mid-point of the chord PQ . The tangents at P and Q meet at the point N .

(i) Show that the equation of the tangent at P is $y = px - ap^2$ 3

(ii) Show that the co-ordinates of M are $(a(p + q), \frac{a(p^2 + q^2)}{2})$ 2

(iii) Show that the co-ordinates of N are $(a(p + q), apq)$. 3

(iv) Show that MN is parallel to the 'y' axis. 1

(v) Find the co-ordinates of T , the mid-point of MN . 2

(vi) Show that T lies on the parabola. 2

SECTION 3

(start a new page)

- 1) By using mathematical induction show that

$$\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$
 where n is a positive integer. 4

- 2) Prove by Mathematical Induction that $3^n + 7^n$ is divisible by 10 for n a positive **odd** number. 4

SECTION 4

(start a new page)

- 1) Solve $\cos 2x = \cos x$ for $0^\circ \leq x \leq 360^\circ$ 3

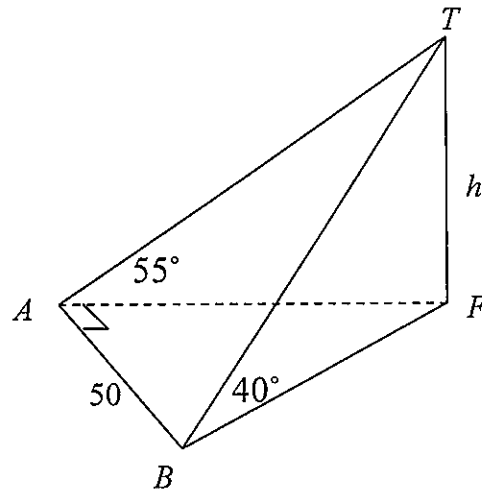
- 2) Express $\sqrt{3} \cos x + \sin x$ in the form $A \cos(x - \alpha)$ and hence find the minimum value of $\sqrt{3} \cos x + \sin x$ and a value of x that gives this minimum value. 3

- 3) If $t = \tan \frac{x}{2}$ write an expression for $\sin x$, $\cos x$ and $\tan x$ in terms of t . Hence solve $\sin x + \cos x = 1$ for $0^\circ \leq x \leq 360^\circ$ 3

- 3) Solve $\sqrt{3} \sin x - \cos x = 1$ for $0^\circ \leq x \leq 360^\circ$. 3

Section 4 continued over the page.

5)



The diagram shows a tower of height h metres standing on level ground. The angle of elevation of the top T of the tower from two points A and B on the ground nearby are 55° and 40° respectively. The distance AB is 50 metres and the interval AB is perpendicular to the interval AF , where F is the foot of the tower.

a) Find AT and BT in terms of h . 2

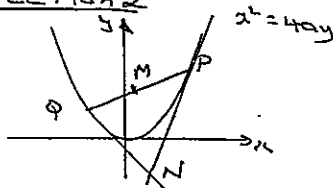
b) Using triangle BAT show that $h = \frac{50 \sin 55^\circ \sin 40^\circ}{\sqrt{\sin^2 55^\circ - \sin^2 40^\circ}}$ 3

c) Hence find the height of the tower, correct to the nearest metre. 1

Section 1

- 1) B
- 2) D
- 3) A
- 4) C

Section 2



i) $x^2 = 4ay$
 $y = \frac{x^2}{4a}$

$\frac{dy}{dx} = \frac{2x}{2a} = \frac{x}{a}$
 at $x = 2ap$: $\frac{dy}{dx} = \frac{2ap}{a} = 2p$

$\therefore$ eqn. of the tangent
 $y - ap^2 = p(x - 2ap)$
 $y - ap^2 = px - 2ap^2$
 $y = px - ap^2$

ii) $M = \left(\frac{2ap + 2ap}{2}, \frac{ap^2 + ap^2}{2} \right)$
 $= \left(\frac{2a(p+q)}{2}, \frac{a(p^2+q^2)}{2} \right)$
 $= \left(a(p+q), \frac{a(p^2+q^2)}{2} \right)$

iii) $y = px - ap^2 \dots (1)$
 $y = qx - aq^2 \dots (2)$
 equating (1) and (2)
 $px - ap^2 = qx - aq^2$

$px - qx = ap^2 - aq^2$
 $x(p - q) = a(p + q)(p - q)$
 $x = a(p + q)$

Sub into (1)
 $y = ap(p + q) - ap^2$
 $y = ap^2 + apq - ap^2$
 $y = apq$
 $\therefore N(a(p + q), apq)$

iv) As M and N have the same x-coordinate, MN must be a vertical line. Therefore parallel to the y-axis.

v) $T = \left(\frac{a(p+q) + a(p+q)}{2}, \frac{apq + apq}{2} \right)$
 $= \left(a(p+q), \frac{2apq + a(p^2+q^2)}{4} \right)$

vi) $x^2 = 4ay$
 $a^2(p+q)^2 = 4a \left(\frac{2apq + a(p^2+q^2)}{4} \right)$

$a^2(p+q)^2 = a^2(2pq + p^2 + q^2)$
 $a^2(p+q)^2 = a^2(p+q)^2$
 $\therefore T$ lies on the parabola.

Section 3

1) $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$

Prove true for $n = 1$.

$\frac{1}{1 \cdot 2} = \frac{1}{1+1}$
 $= \frac{1}{2} = \frac{1}{2}$ True.

Assume true for $n = k$ (where k is a positive integer)

i.e. $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \dots + \frac{1}{k(k+1)} = \frac{k}{k+1}$

Prove true for $n = k+1$

i.e. $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)} = \frac{k+1}{k+2}$

Proof:

L.H.S. $= \frac{k}{k+1} + \frac{1}{(k+1)(k+2)}$ From the assumption
 $= \frac{k(k+2) + 1}{(k+1)(k+2)}$
 $= \frac{k^2 + 2k + 1}{(k+1)(k+2)}$
 $= \frac{(k+1)^2}{(k+1)(k+2)}$
 $= \frac{k+1}{k+2}$
 $= R.H.S.$

Therefore the statement is true for $n = k+1$ assuming true for $n = k$. Therefore it is true by Mathematical Induction.

2) $3^n + 7^n$ is divisible by 10 n odd.
 Prove true for $n = 1$.

i.e. $3^1 + 7^1$ is divisible by 10 - True.

Assume true for $n = k$; k odd.

i.e. $3^k + 7^k = 10M$, M a positive integer

Prove true for $n = k+2$.

i.e. $3^{k+2} + 7^{k+2}$ is divisible by 10

$$\begin{aligned}
 &= 9 \cdot 3^k + 49 \cdot 7^k \\
 &= 49 \cdot 3^k + 49 \cdot 7^k - 40 \cdot 3^k \\
 &= 49(3^k + 7^k) - 40 \cdot 3^k \\
 &= 49 \times 10M - 40 \cdot 3^k \quad \text{from assumption} \\
 &= 10(49M - 4 \cdot 3^k) \quad \text{which is divisible by 10.}
 \end{aligned}$$

Therefore the statement is true for $n = k+1$ assuming true for $n = k$. Therefore true by mathematical induction.

SECTION 4

$$\cos 2x = \cos x$$

$$\begin{aligned}
 1) \quad &\cos 2x - \cos x = 0 \\
 &2\cos^2 x - 1 - \cos x = 0 \\
 &2\cos^2 x - \cos x - 1 = 0 \\
 &(2\cos x + 1)(\cos x - 1) = 0 \\
 &\cos x = -\frac{1}{2} \quad \cos x = 1 \\
 &x = 120^\circ, 240^\circ, 0^\circ, 360^\circ
 \end{aligned}$$

$$\begin{aligned}
 2) \quad &\sqrt{3}\cos x + \sin x = A\cos(x-\alpha) \\
 &= A\cos x \cos \alpha + A\sin x \sin \alpha
 \end{aligned}$$

$$\begin{aligned}
 &\text{equating coefficients} \\
 &A\cos \alpha = \sqrt{3} \quad \dots (1) \\
 &A\sin \alpha = 1 \quad \dots (2) \\
 (2) \div (1) \quad \tan \alpha &= \frac{1}{\sqrt{3}} \\
 \alpha &= 30^\circ
 \end{aligned}$$

$$\begin{aligned}
 (1)^2 + (2)^2 \quad A^2(\sin^2 \alpha + \cos^2 \alpha) &= 4 \\
 A^2 &= 4 \\
 A &= 2
 \end{aligned}$$

$$\begin{aligned}
 \therefore \sqrt{3}\cos x + \sin x &= 2\cos(x-30^\circ) \\
 \therefore \text{max value} &= 2 \\
 \text{when } x &= 210^\circ
 \end{aligned}$$

$$3) \sin x = \frac{2t}{1+t^2}$$

$$\cos x = \frac{1-t^2}{1+t^2}$$

$$\tan x = \frac{2t}{1-t^2}$$

$$\sin x + \cos x = 1$$

$$\frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2} = 1$$

$$2t + 1 - t^2 = 1 + t^2$$

$$2t^2 - 2t = 0$$

$$2t(t-1) = 0$$

$$t=0, \quad t=1$$

$$\tan \frac{x}{2} = 0 \quad \tan \frac{x}{2} = 1$$

$$\frac{x}{2} = 0, 180^\circ, 360^\circ; \quad \frac{x}{2} = 45^\circ, 225^\circ$$

$$\begin{aligned}
 x &= 0, 360^\circ, 90^\circ \text{ for } 0 \leq x \leq 360^\circ \\
 \text{check } x &< 180^\circ
 \end{aligned}$$

$$\sin 180 + \cos 180 = 1$$

$$0 - 1 = -1$$

false $\therefore$ not a solution

$$\begin{aligned}
 4) \quad &\sqrt{3}\sin x - \cos x = 1 \\
 \text{let } \sqrt{3}\sin x - \cos x &= R\sin(x-\alpha) \\
 &= R\sin x \cos \alpha - R\cos x \sin \alpha
 \end{aligned}$$

$$\text{equating coefficients} \quad R\cos \alpha = \sqrt{3} \quad \dots (1)$$

$$R\sin \alpha = 1 \quad \dots (2)$$

$$(2) \div (1) \quad \tan \alpha = \frac{1}{\sqrt{3}}$$

$$\alpha = 30^\circ$$

$$(1)^2 + (2)^2: R^2(\sin^2 \alpha + \cos^2 \alpha) = 4$$

$$R^2 = 4$$

$$R = 2$$

$$\therefore \sqrt{3}\sin x - \cos x = 2\sin(x-30^\circ)$$

$$\therefore \sqrt{3}\sin x - \cos x = 1$$

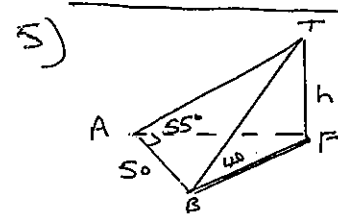
$$1 = 2\sin(x-30^\circ) = 1$$

$$\sin(x-30^\circ) = \frac{1}{2}$$

$$x-30^\circ = 30^\circ, 150^\circ$$

$$x = 60^\circ, 180^\circ$$

N.B. 't' method could also be used.



a) from $\triangle ATF$

$$\frac{h}{AT} = \sin 55^\circ$$

$$AT = \frac{h}{\sin 55^\circ}$$

from $\triangle BTF$

$$\frac{h}{TB} = \sin 40^\circ$$

$$TB = \frac{h}{\sin 40^\circ}$$

b) $\triangle BAT$ is right angled
 $\therefore TB^2 = AT^2 + AB^2$ (Pythagoras)

$$\frac{h^2}{\sin^2 40^\circ} = \frac{h^2}{\sin^2 55^\circ} + 50^2$$

$$\frac{h^2}{\sin^2 40^\circ} - \frac{h^2}{\sin^2 55^\circ} = 2500$$

$$h^2 \left(\frac{1}{\sin^2 40^\circ} - \frac{1}{\sin^2 55^\circ} \right) = 2500$$

$$h^2 \left(\frac{\sin^2 55^\circ - \sin^2 40^\circ}{\sin^2 55^\circ \sin^2 40^\circ} \right) = 2500$$

$$h^2 = \frac{2500 \sin^2 55^\circ \sin^2 40^\circ}{\sin^2 55^\circ - \sin^2 40^\circ}$$

$$h = \frac{50 \sin 55^\circ \sin 40^\circ}{\sqrt{\sin^2 55^\circ - \sin^2 40^\circ}}$$

c) 52 metres.

SECTION I

$$\begin{aligned} iz &= i(3-i) \\ &= 3i - i^2 \\ &= 1+3i \end{aligned}$$

$$\therefore \bar{iz} = 1-3i \quad \text{Hence C}$$

$$\text{If } \operatorname{Re}(z) = 2$$

$$x = 2 \quad \checkmark$$

$$\text{If } |z| = |2-4i|$$

$$x = 2 \quad \checkmark$$

$$\text{If } z + \bar{z} = 4$$

$$x+iy + x-iy = 4$$

$$2x = 4$$

$$x = 2 \quad \checkmark$$

Hence C

$$\text{If } w = 1+i$$

$$\frac{1}{w} = \frac{1}{1+i} \times \frac{1-i}{1-i}$$

$$= \frac{1-i}{2}$$

$$= \frac{1}{2} - \frac{1}{2}i$$

Hence A

$$4. \frac{z_1}{z_2} = \frac{2}{4} \operatorname{cis} \left(\frac{5\pi}{6} - \frac{\pi}{6} \right)$$

$$= \frac{1}{2} \operatorname{cis} \frac{4\pi}{6}$$

$$= \frac{1}{2} \operatorname{cis} \frac{2\pi}{3}$$

Hence A

SECTION II

$$1. (i) 2z + i$$

$$= 2(3-2i) + i(1-i)$$

$$= 6-4i + i-i^2$$

$$= 7-3i$$

(1)

$$(ii) \bar{z}w$$

$$= (3+2i)(1-i)$$

$$= 3-3i+2i-2i^2$$

$$= 5-i$$

(1)

$$(iii) \frac{4}{w}$$

$$= \frac{4}{1-i} \times \frac{1+i}{1+i}$$

$$= \frac{4(1+i)}{1-i^2}$$

$$= \frac{4(1+i)}{2}$$

$$= 2+2i$$

(2)

$$(iv) \left| \frac{4}{w} \right|$$

$$= \sqrt{(2)^2 + (-2)^2} \quad \text{since } \left(\frac{4}{w} \right) = 2-2i$$

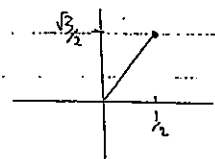
$$= \sqrt{8}$$

$$= 2\sqrt{2}$$

(1)

$$2. (i) \frac{1+i\sqrt{3}}{2}$$

$$= \frac{1}{2} + \frac{\sqrt{3}}{2}i$$



$$\therefore r = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}$$

$$= 1$$

$$\theta = \tan^{-1} \left(\frac{\sqrt{3}/2}{1/2} \right)$$

$$= \tan^{-1} (\sqrt{3})$$

$$= \frac{\pi}{3}$$

$$\therefore \frac{1+i\sqrt{3}}{2} = \operatorname{cis} \frac{\pi}{3} \quad (2)$$

$$(ii) \text{ If } z = \operatorname{cis} \frac{\pi}{3}$$

$$z^3 = \operatorname{cis} 3\left(\frac{\pi}{3}\right)$$

$$z^3 = \operatorname{cis} \pi$$

$$= \cos \pi + i \sin \pi$$

$$= -1 + i \cdot 0$$

$$= -1$$

(1)

$$(iii) z^7 = z^6 \cdot z$$

$$= (z^3)^2 \cdot z$$

$$= (-1)^2 \cdot z$$

$$= z$$

$$= \frac{1+i\sqrt{3}}{2} \quad \text{or} \quad \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

(1)

$$3. (i) \text{ Let } z = \cos \theta + i \sin \theta$$

$$\text{If } z^5 = -1$$

$$\cos 5\theta + i \sin 5\theta = -1 + i \cdot 0$$

$$\therefore \cos 5\theta = -1$$

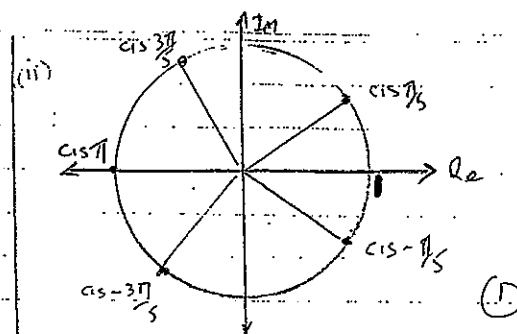
$$5\theta = \pi, 3\pi, 5\pi, 7\pi, 9\pi$$

$$\theta = \frac{\pi}{5}, \frac{3\pi}{5}, \pi, \frac{7\pi}{5}, \frac{9\pi}{5}$$

$$\text{ie } \theta = \frac{\pi}{5}, \frac{3\pi}{5}, \pi, \frac{7\pi}{5}, \frac{9\pi}{5} \quad (2)$$

$$\text{since } \frac{7\pi}{5} = \frac{-3\pi}{5} \quad \text{and} \quad \frac{9\pi}{5} = \frac{-\pi}{5}$$

$$\text{The roots are } \operatorname{cis} \frac{\pi}{5}, \operatorname{cis} \frac{3\pi}{5}, \operatorname{cis} \pi, \operatorname{cis} \frac{7\pi}{5}, \operatorname{cis} \frac{9\pi}{5}$$



SECTION II

$$1. (i) \text{ Let } \sqrt{-6i} = x+iy$$

$$0-6i = (x+iy)^2$$

$$0-6i = x^2 - y^2 + 2xyi$$

$$x^2 - y^2 = 0, \quad 2xy = -6$$

$$y = \frac{-3}{x}$$

$$\therefore x^2 - \left(\frac{-3}{x}\right)^2 = 0$$

$$x^2 - \frac{9}{x^2} = 0$$

$$x^4 - 9 = 0$$

$$(x^2-3)(x^2+3) = 0$$

$$x = \pm \sqrt{3}$$

$$y = \mp \frac{3}{\sqrt{3}}$$

$$y = \mp \sqrt{3}$$

(2)

$$\therefore \sqrt{-6i} = \sqrt{3} - i\sqrt{3} \quad \text{or} \quad -\sqrt{3} + i\sqrt{3}$$

$$(ii) \text{ If } z^2 + (1+i)z + 2i = 0$$

$$z = \frac{-(1+i) \pm \sqrt{(1+i)^2 - 8i}}{2}$$

$$= \frac{-(1+i) \pm \sqrt{1+2i+i^2-8i}}{2}$$

$$= \frac{-(1+i) \pm \sqrt{-7-6i}}{2}$$

$$= \frac{-(1+i) \pm \sqrt{-6i}}{2}$$

$$= \frac{-(1+i) \pm (\sqrt{3}-i\sqrt{3})}{2}$$

$$= \frac{(\sqrt{3}-1)}{2} - \frac{(\sqrt{3}+1)i}{2} \quad (2)$$

$$\text{or } -\frac{(\sqrt{3}+1)}{2} + \frac{(\sqrt{3}-1)i}{2}$$

$$(ii) \frac{5\pi}{12} = \frac{\pi}{6} + \frac{\pi}{4}$$

$$\therefore \tan \frac{5\pi}{12} = \tan \left(\frac{\pi}{6} + \frac{\pi}{4} \right)$$

$$= \frac{\tan \frac{\pi}{6} + \tan \frac{\pi}{4}}{1 - \tan \frac{\pi}{6} \tan \frac{\pi}{4}}$$

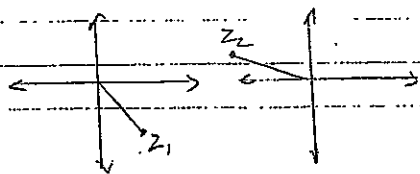
$$= \frac{\frac{1}{\sqrt{3}} + 1}{1 - \frac{1}{\sqrt{3}}}$$

$$= \frac{\sqrt{3}+1}{\sqrt{3}-1}$$

$$= \frac{\sqrt{3}+1}{\sqrt{3}-1}$$

$$(iv) \text{ Let } z_1 = \frac{(\sqrt{3}-1)}{2} - \frac{(\sqrt{3}+1)i}{2}$$

$$z_2 = -\frac{(\sqrt{3}+1)}{2} + \frac{(\sqrt{3}-1)i}{2}$$



$$\arg z_1 = \tan^{-1} \left[\frac{-(\sqrt{3}+1)}{\frac{(\sqrt{3}-1)}{2}} \right]$$

$$= -\tan^{-1} \left[\frac{\sqrt{3}+1}{\sqrt{3}-1} \right]$$

$$= -\frac{5\pi}{12}$$

$$\arg(z_2) = \pi - \tan^{-1} \left[\frac{(\sqrt{3}-1)}{\frac{(\sqrt{3}+1)}{2}} \right]$$

$$= \pi - \frac{\pi}{12}$$

$$= \frac{11\pi}{12}$$

$$\therefore \arg(z_1) + \arg(z_2) = -\frac{5\pi}{12} + \frac{11\pi}{12}$$

$$= \frac{\pi}{2}$$

$$2. \text{ Let } z = x+iy$$

$$z-A = x+iy - (1-i)$$

$$= x-1 + i(y+1)$$

$$z-B = x+iy - (2+i)$$

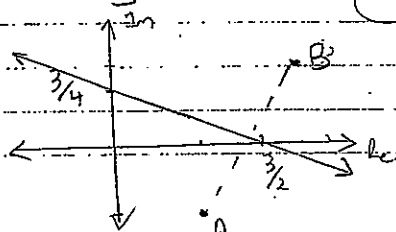
$$= x-2 + i(y-1)$$

$$\text{If } |z-A| = |z-B|$$

$$\sqrt{(x-1)^2 + (y+1)^2} = \sqrt{(x-2)^2 + (y-1)^2}$$

$$\therefore x^2 - 2x + 1 + y^2 + 2y + 1 = x^2 - 4x + 4 + y^2 - 2y + 1$$

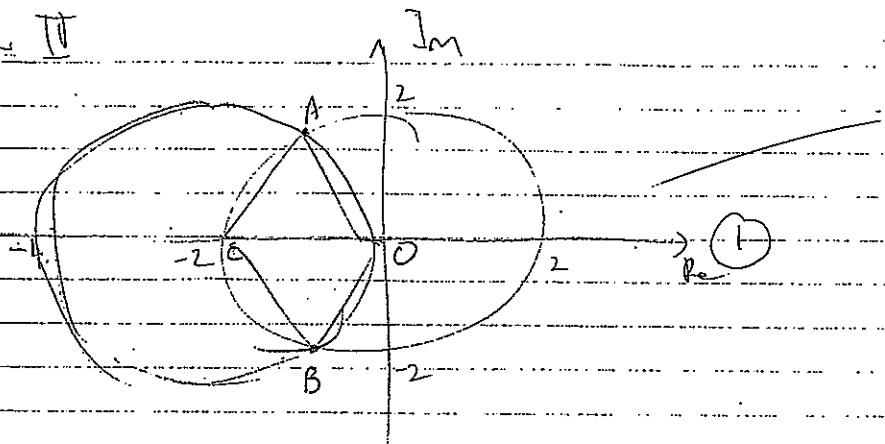
$$\text{i.e. } 2x + 4y - 3 = 0 \quad (3)$$



N.B. The locus is the perpendicular bisector of AB

Solution II

1/ (i)

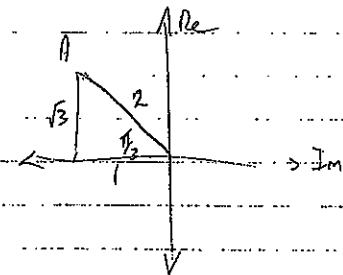


(ii) Let the points of intersection be A & B
Let the center of the circle be O = C

$\triangle AOC$ & $\triangle BOC$ are equilateral since $OA = OC = OB = OC = BC$
 $= 2 \text{ units}$

$\therefore A$ is the point $\cos \frac{2\pi}{3}$

and B is the point $\cos -\frac{2\pi}{3}$



$\therefore A = -1 + \sqrt{3}i$

by symmetry
 $B = -1 - \sqrt{3}i$

$$2/ (i) \text{ If } z = \cos \theta + i \sin \theta$$

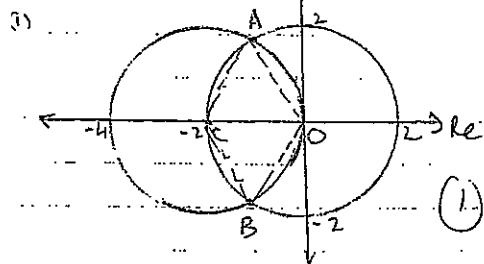
$$z^n = \cos(n\theta) + i \sin(n\theta)$$

$$z^{-n} = \cos(-n\theta) + i \sin(-n\theta)$$

$$= \cos(n\theta) - i \sin(n\theta)$$

$$\therefore z^n + z^{-n} = 2 \cos n\theta$$

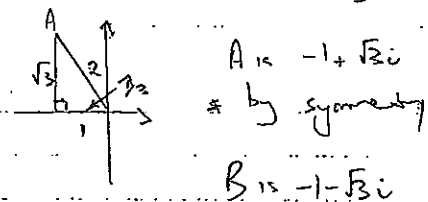
QUESTION IV



iii) Let the points of intersection be A & B and the centres be O & C
 $OA = OB = OC = CA = CB = 2 \text{ units}$
 $\therefore \Delta ACC$ is equilateral

$\therefore A$ is the point $\text{cis } \frac{2\pi}{3}$

$\therefore B$ is the point $\text{cis } -\frac{2\pi}{3}$



$$2. (i) z^n = \cos(n\theta) + i \sin(n\theta)$$

$$z^{-n} = \cos(-n\theta) + i \sin(-n\theta)$$

$$= \cos(n\theta) - i \sin(n\theta)$$

$$\therefore z^n + z^{-n} = 2 \cos(n\theta) \quad (1)$$

$$(ii) \text{ If } z = \cos \theta + i \sin \theta$$

$$z^n - z^{-n} = 2i \sin(n\theta)$$

$$\therefore z - \frac{1}{z} = 2i \sin \theta$$

$$\text{Now } \left(z - \frac{1}{z}\right)^3 = (2i \sin \theta)^3$$

$$= -8i \sin^3 \theta$$

$$\therefore -8i \sin^3 \theta = z^3 - 3z^2 \cdot \frac{1}{z} + 3z \cdot \frac{1}{z^2} - \frac{1}{z^3}$$

$$-8i \sin^3 \theta = \left(z^3 - \frac{1}{z^3}\right) - 3\left(z - \frac{1}{z}\right)$$

$$-8i \sin^3 \theta = 2i \sin 3\theta - 3(2i \sin \theta)$$

$$-8i \sin^3 \theta = 2i \sin 3\theta - 6i \sin \theta$$

$$\therefore \sin^3 \theta = \frac{2i \sin 3\theta - 6i \sin \theta}{-8i}$$

$$= -\frac{\sin 3\theta}{4} + \frac{3 \sin \theta}{4}$$

$$= \frac{3 \sin \theta}{4} - \frac{\sin 3\theta}{4} \quad (3)$$

$$3. (i) \text{ If } z^3 - 1 = 0$$

$$z^3 + 0z^2 + 0z - 1 = 0$$

The sum of the roots is $-\frac{b}{a}$

$$\therefore 1 + \omega + \omega^2 = 0 \quad (1)$$

$$(ii) (z - \omega)(z - \omega^2)$$

$$= z^2 - z\omega^2 - z\omega + \omega^3$$

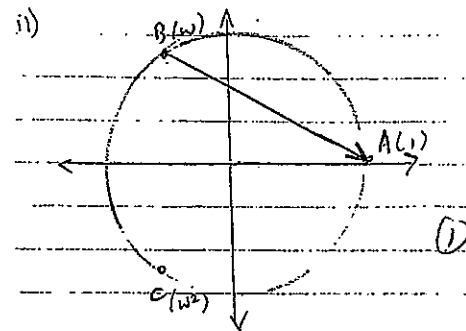
$$= z^2 - z(\omega^2 + \omega) + \omega^3$$

$$= z^2 - 2(-1) + 1$$

since $\omega^2 + \omega = -1$ & ω is a root of $z^3 - 1 = 0$, $\omega^3 = 1$

$$\therefore (z - \omega)(z - \omega^2) = z^2 + z + 1 \quad (2)$$

ii)



$$iv) \frac{\overrightarrow{BA}}{\overrightarrow{CA}} = \frac{(1 - \omega)(1 - \omega^2)}{(1 - \omega^2)(1 - \omega)}$$

$$= \frac{1 - \omega^2 - \omega + \omega^3}{1 - \omega^2 - \omega + \omega^3}$$

$$= \frac{1 - (\omega^2 + \omega) + \omega^3}{1 - (\omega^2 + \omega) + \omega^3}$$

$$= \frac{1 - (-1) + 1}{1 - (-1) + 1}$$

$$= 3 \quad (1)$$